مؤسسة الإمارات للتعليم المدرسي
EMIRATES SCHOOLS ESTABLISHMENT

CHEMISTRY



أسئلة مراجعة

Revision questions

11A



مؤسسة الإمارات للتعليم المدرسي
EMIRATES SCHOOLS ESTABLISHMENT

EoT2 Exam Coverage

11A



Academic Year العام الدراسي	2024/2025
Term الفصل	1
Subject المادة	Chemistry /Inspire الكيمياء / انسباير
Plan M101-A	
Grade الصف	11
Stream المسار	Advanced المتقدم

مؤسسة الإمارات
للتعليم المدرسي
EMIRATES SCHOOLS
ESTABLISHMENT

Unit:

Energy and chemical Change



11A

Calculate the energy in joules in the below picture	احسب مقدار الطاقة بوحدة الجول (J) في الصورة أدناه؟
245 Calories 	
A. 1.03×10^6	
B. 1.03×10^3	
C. 2.45×10^5	
D. 5.86×10^4	

Calculate the energy in joules in the below picture	احسب مقدار الطاقة بوحدة الجول (J) في الصورة أدناه؟
440 Cal 	
A. 1.84×10^6	
B. 1.84×10^3	
C. 1.1×10^3	
D. 4.4×10^8	

How much energy in joules is supplied by a breakfast bar containing 240 nutritional calories?	ما مقدار الطاقة بالجول التي توفرها وجبة الإفطار التي تحتوي على **240 Cal** ؟
A. 240 J	
B. 1.00×10^6 J	
C. 5.74×10^4 J	
D. 1004 J	

7) An exothermic reaction release 86.5kJ, How many kcal of energy are released ?

يطلق تفاعل طارد للحرارة 86.5kJ من الحرارة . كم مقدار الطاقة بوحدة kcal.

A. 20.67 kcal

B.20674 kcal

C. 361.92 kcal

D. 361916 kcal

1 cal = 4.184 J
1 kJ = 1000 J
1 kcal = 1000 cal

The same amount of heat is added to a 10g sample of each of the following metals . If each metal is initially at 20°C , which metal will reach the highest temperature ?

أضيف ت نفس كمية الحرارة إلى 10g من كل عينة من الفلزات التالية . إذا كانت درجة الحرارة الابتدائية لكل فلز هي 20°C ، أي الفلزات سوف يصل أعلى درجة حرارة ؟

A. Calcium
B. Beryllium
C. Gold
D. Copper

Metal الفلز	Specific Heat الحرارة النوعية
Beryllium البيريليوم	1.82 J/(g°C)
Calcium الكالسيوم	0.653 J/(g°C)
Copper النحاس	0.385 J/(g°C)
Gold الذهب	0.129 J/(g°C)

A 355g sample of unknown substance was heated from 22.4 °C to 43.6 °C the substance absorbs 6.75 KJ of energy.
Using the table below , which is the substance ?

Substance	الذهب Gold	الفضة Silver	الألمنيوم Aluminum	الحديد Iron
Specific heat	0.129	0.235	0.897	0.449

A. Iron

B. Aluminum

C. Silver

D. Gold

A 25.0–g of an alloy absorbed 250 joules of heat as its temperature changed from 25.0°C to 78.0°C. What is the specific heat of the alloy? سبيكة كتلتها 25.0g امتصت حرارة مقدراها 250J وتغيرت درجة الحرارة من 25.0°C إلى 78.0°C ، ما الحرارة النوعية للسبيكة ؟
A. 0.84 J /(g. °C)
B.0.189 J /(g. °C)
C. 0.129 J /(g. °C)
D. 0.449 J /(g. °C)

A 155 g sample of unknown substance was heated from 25.0°C to 40.0°C the substance piece absorbed 5696J of energy . What is the specific heat of the substance ? تم تسخين عينة كتلتها 155g من مادة غير معلومة من 25.0°C إلى 40.0°C وامتصت هذه المادة خلال العملية 5696J من الطاقة. ما الحرارة النوعية لهذه المادة؟
A. 2.03 J /(g. °C)
B.2.45 J /(g. °C)
C. 4.18 J /(g. °C)
D. 235 /(g. °C)

Two pieces of Aluminum and iron were left to sit in the sun at the same time and for the same length of time. What is the mass of piece of iron (g) whose tempertaure increase by the same amount as piece of aluminum ?

الحديد Iron	الألمنيوم Aluminium	المادة Substance
.....	47.0 g	Mass الكتلة
0.449	0.897	الحرارة النوعية Specific Heat J/(g. ^{0}C)
30.0^0 C	30.0^0 C	ΔT

A. 93.9

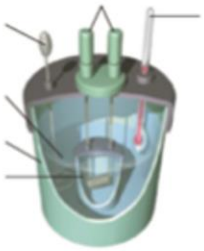
B. 45.6

C. 35.5

D. 53.5

Which of the following statements is incorrect regarding the two calorimeters in the figure below?

أي العبارات التالية غير صحيحة فيما يتعلق بالمُسعرين في الشكل أدناه؟

Calorimeter (2) المسعر	Calorimeter (1) المسعر
	

A. The data to be collected in both calorimeters is the change in temperature of the mass of water

تتمثل البيانات التي سيتم جمعها في كلا المسعرين في تغير درجة حرارة كمية الماء

B. All reactions carried out in calorimeter (2) occur at constant pressure

جميع التفاعلات التي تحدث بداخل المُسعر (2) تحدث تحت ضغط ثابت

C. A low – friction stirrer in calorimeter (1) ensure uniform temperature

يضمن المحرك منخفض الاحتكاك في المُسعر (1) درجة حرارة موحدة

D. Only calorimeter (1) can be used to determine specific heat of a metal

يُمكن استخدام المُسعر (1) فقط في تحديد الحرارة النوعية لفلز

In which of the following cases is it impossible to measure or impractical to measure ΔH of rection using a calorimeter?

في أي الحالات التالية يصبح من المستحيل أو من غير العملي قياس ΔH في تفاعل باستخدام المسعر الحراري ؟

عند حدوث التفاعل ببطء شديد As the reaction occurs so slowly	I
عندما تنتج نواتج غير مرغوب فيها When unwanted products are produced	II
عند حدوث التفاعل في ظل ظروف يصعب تكرارها في المختبر As the reaction occurs under conditions difficult to duplicate in a laboratory	III

A. I only

B. II only

C. I, III only

D. I, II and III

In which of the following cases is it impossible to measure or impractical to measure ΔH of rection using a calorimeter?

في أي الحالات التالية يصبح من المستحيل أو من غير العملي قياس ΔH في تفاعل باستخدام المسعر الحراري ؟

عند حدوث التفاعل ببطء شديد As the reaction occurs so slowly	I
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عند حدوث التفاعل في ظل ظروف يصعب تكرارها في المختبر As the reaction occurs under conditions difficult to duplicate in a laboratory	III

A. I only

B. II only

C. I, III only

D. I, II and III

When a **88.2 g** piece of hot alloy is placed in **175 g** of cold water in a calorimeter, the temperature of the alloy decreases by **76.4 ^{0}C,** while the temperature of the water increases by **15.6 ^{0}C** What is the specific heat of the alloy **J/ (g.^{0}C)**?

عند وضع قطعة من سبيكة ساخنة كتلتها **88.2 g** في **175 g** من الماء البارد في كالوريميتر . تقل درجة حرارة السبيكة بمقدار **76.4 ^{0}C** ، بينما ترتفع درجة حرارة الماء بمقدار **15.6 ^{0}C** ما الحرارة النوعية للسبيكة بوحدة **J/ (g. ^{0}C)**؟

The specific heat of water = 4.184 J/(g. ^{0}C)

الحرارة النوعية للماء =4.184 J/(g. ^{0}C)

A. 1.7
B. 2.4
C. 0.809
D. 0.129

What is the amount of heat absorbed when 5.50 g of iron is heated from 25.0°C to 95.0°C? (Specific heat of iron= 0.449 J/(g.°C))	ما مقدار الحرارة التي يتم امتصاصها عندما يتم تسخين 5.50 g من الحديد من درجة حرارة 25.0°C إلى 95.0°C؟ (الحرارة النوعية للحديد= 0.449 J/(g.°C))
A. 235J	
B. 62.0 J	
C. 385 J	
D. 173 J	

If 250 g of ethanol at 75.0°C loses 4655 J of heat, what is the final temperature of ethanol ? (specific Heat of ethanol : 2.44 J/(g. °C) إذا 250 g من الإيثانول درجة حرارته 75.0°C كمية من الحرارة 4655 J . فما درجة الحرارة النهائية للإيثانول ؟ (الحرارة النوعية للإيثانول هي 2.44 J/(g. °C
A. 82.6 °C
B. 67.4 °C
C. 59.5 °C
D. 45.8 °C

Which pf the data in the following table is correct?

أي من البيانات الواردة في الجدول التالي صحيحة؟

الرقم Number	العملية Process	التغير في المحتوى الحراري The change in enthalpy	إشارة التغير في المحتوى الحراري The sign of the enthalpy change
1	$C_2H_5OH(s) \rightarrow C_2H_5OH(l)$	طاردة للحرارة exothermic	موجبة positive
2	$NH_3(l) \rightarrow NH_3(s)$	ماصة للحرارة endothermic	موجبة positive
3	$CH_3OH(l) \rightarrow CH_3OH(g)$	ماصة للحرارة endothermic	سالبة negative
4	$H_2O(g) \rightarrow H_2O(l)$	طاردة للحرارة exothermic	سالبة negative

A. 1 only

B. 1,3

C. 4 only

D. 3,4

Regarding to the reaction **A → C** which of the following represented by the diagram below?

بالنسبة للتفاعل **A → C** أي مما يلي يمثله الشكل أدناه ؟



A.	$H_{\text{متفاعلات}} < H_{\text{نواتج}}$ (reactans / products)
B.	$H_{\text{نواتج}} = H_{\text{متفاعلات}}$ (products / reactans)
C.	$H_{\text{متفاعلات}} > H_{\text{نواتج}}$ (reactans / products)
D.	$\Delta H > 0$

Which of the following is **correct** about hot pack ?

أي مما يلي **صحيح** حول الكمادة الساخنة ؟



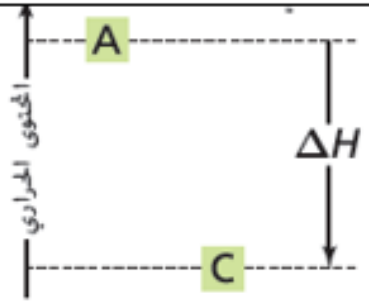
A. Quantity of heat is absorbed from the surrounding	تمتص كمية من الحرارة من المحيط إلى الكمادة الساخنة
B. The value of ΔH_{rxn} for reaction is a negative	إشارة ΔH_{rxn} للتفاعل قيمة سالبة
C. The reaction is endothermic	التفاعل ماص للحرارة
D. Heat transfer from a person leg to the hot pack	تنتقل الحرارة من ساق الشخص إلى الكمادة الساخنة

Which of the following is incorrect about cold pack ?

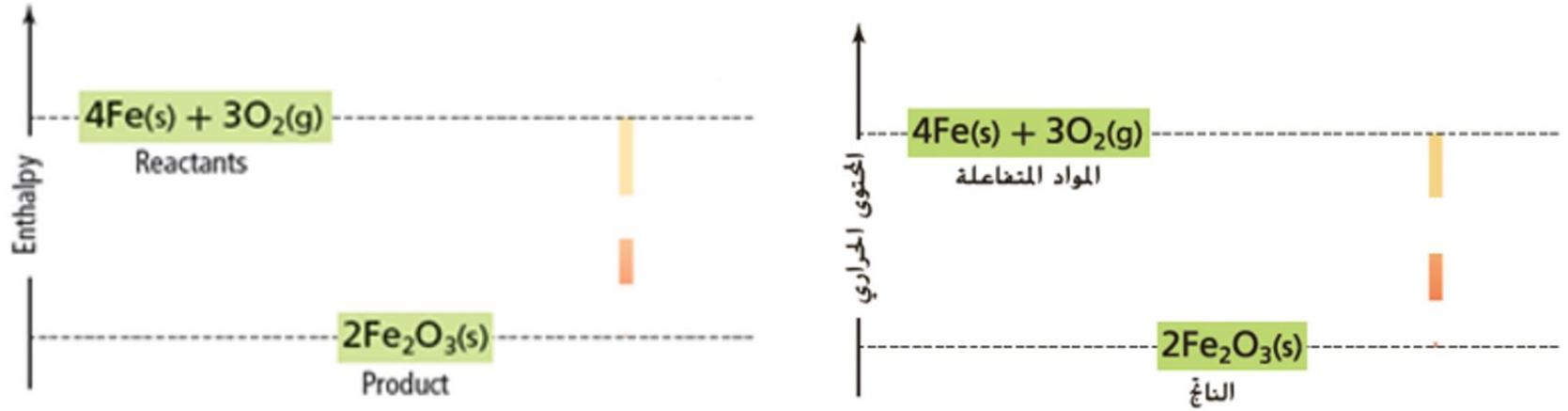
أي مما يلي غير صحيح حول الكمادة الباردة ؟



A. Quantity of heat is absorbed from the surrounding	تمتص كمية من الحرارة من المحيط إلى الكمادة الباردة
B. The value of ΔH_{rxn} for reaction is a positive	إشارة ΔH_{rxn} للتفاعل قيمة موجبة
C. The reaction is exothermic	التفاعل طارد للحرارة
D. Heat transfer from a person leg to the cold pack	تنتقل الحرارة من ساق الشخص إلى الكمادة الباردة

In the below figure. Which of the following is true?	في الشكل أدناه. أي العبارات التالية صحيحة؟
	
A. The reaction is endothermic	A. التفاعل ماص للحراره
B. The sign of ΔH is positive	B. إشاره ΔH موجبة
C. the product (A) has a lower energy than the reactant (C).	C. طاقة المواد A أقل من طاقة المواد C
D. the product (C) has a lower energy than the reactant (A).	D. طاقة المواد C أقل من طاقة المواد A

Which of the following is true regarding the figure below



A. The reaction is endothermic

B. This reaction is used in the Cold-Pack

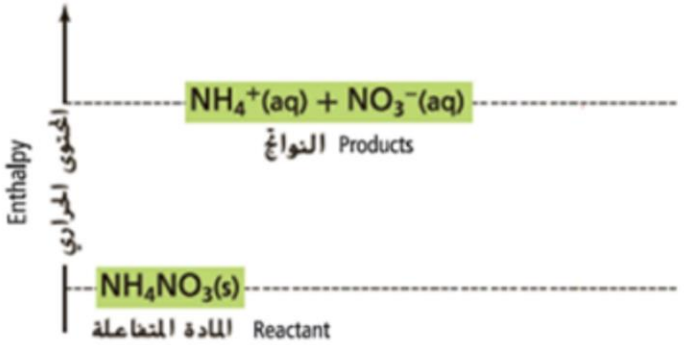
C. This reaction is used in the Heat-Pack

D. The sign of enthalpy is positive

Which of the following is correct regarding the diagram and figures (1) and (2) below?

أي مما يأتي صحيح فيما يتعلق بالرسم البياني والشكلين (1) و (2) أدناه؟

Figure (2) الشكل	Figure (1) الشكل
	



A. The graph shows that the reaction is exothermic and is used to warm cold hands in figure (1)

يُوضح الرسم البياني أن التفاعل طارد للحرارة ويُستخدم لتدفئة اليدين الباردتين في الشكل (1)

B. The graph shows that the reaction is endothermic and used to warm cold hands in figure (1)

يُوضح الرسم البياني أن التفاعل ماص للحرارة ويُستخدم لتدفئة اليدين الباردتين في الشكل (1)

C. The graph shows that the reaction is endothermic and used to cool the leg of a person in figure (2)

يُوضح الرسم البياني أن التفاعل ماص للحرارة ويُستخدم لتبريد ساق الشخص في الشكل (2)

D. The graph shows that the reaction is exothermic and is used to cool the leg of a person in figure (2)

يُوضح الرسم البياني أن التفاعل طارد للحرارة ويُستخدم لتبريد ساق الشخص في الشكل (2)

What does the zero superscript in the symbol ΔH°_{comb} indicates

ما الذي يشير إليه الصفر العلوي بالرمز ΔH°_{comb}؟

A.	Standard conditions at 1 atm pressure and 298 k(25°C)	الظروف القياسية في ضغط 1 atm ودرجة حرارة 298 k (25°C)
B.	Standard temperature and pressure (STP)	الضغط ودرجة الحرارة القياسيين (STP)
C.	Low pressure and Low temperature	الضغط المنخفض ودرجة حرارة المنخفضة
D.	Conditions are 0 atm pressure and 273 k (0°C)	الظروف في ضغط 0 atm ودرجة حرارة 273 k (0°C)

How much heat is evolves when 206 g of methanol condense to a liquid at its boiling point ?

ما كمية الحرارة المنطلقة عن تكثف 206 g من الميثانول إلى سائل عند درجة غليانه ؟

substance المادة	formula الصيغة	ΔH°_{vap} (KJ/mol)	Molar mass الكتلة المولية
methanol الميثانول	CH_3OH	35.2	32g/mol

A. 187.3 kJ

B. 5.47 kJ

C. 226.6 kJ

D. $2.3 * 10^5$ kJ

Regarding the following equations, which of the following statements is incorrect ?

فيما يتعلق بالمعادلات التالية ، أي العبارات التالية غير صحيحة؟

1	$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} \qquad \Delta H_{comb} = -891\ kJ/mol$
2	$C_8H_{18(l)} + \frac{25}{2}O_{2(g)} \rightarrow 8CO_{2(g)} + 9H_2O_{(l)} + 5471\ kJ$
3	$C_6H_{12}O_{6(s)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)} \qquad \Delta H_{comb} = -2808\ kJ/mol$
4	$H_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow H_2O_{(l)} + 286\ kJ$

A. All equations represented combustion reactions — جميع المعدلات تمثل تفاعلات احتراق

B. All equations represent exothermic reactions — جميع المعادلات تمثل تفاعلات طاردة للحرارة

C. The combustion of one mole of octane **C_8H_{18}** produces more heat than the amount produce by the combustion of one mole of glucose **$C_6H_{12}O_6$**

احتراق مول واحد من الأوكتان **C_8H_{18}** ينتج كمية من الحرارة أكبر من الكمية التي ينتجها احتراق مول واحد من الجلوكوز **$C_6H_{12}O_6$**

D. The heat of combustion of hydrogen **H_2** is + 286 KJ/ mol — حرارة احتراق الهيدروجين **H_2** تساوي +286 KJ/mol

How much heat is liberated by the combustion of 206 g of hydrogen gas?(molar mass of H_2 = 2.01 g/mol ΔH_{comb} = -286 kJ/mol ما كمية الحرارة الناتجة من احتراق 206 g من الهيدروجين ؟ (الكتلة المولية للهيدروجين = 2.01 g/mol)
A. 29,311 kJ
B. 118421 kJ
C. 58916 kJ
D. 572 kJ

How much heat is evolved when 27.o g of glucose ($C_6H_{12}O_6$) is burned according to equation below ?(molar mass of $C_6H_{12}O_6$= 180.18 g/mol $$C_6H_{12}O_{6(s)} + 6O_{2(g)} \rightarrow 6CO_{2(g)} + 6H_2O_{(l)} \quad \Delta H_{comb} = -2808\ kJ$$
A. 136 kJ
B.210 kJ
C. 421 kJ
D.180 kJ

What mass of methane CH_4 (by grams) must burned in order to liberate **3564 kJ** of heat?

(Molar mass of CH_4 =16 g/mol)

ما كتلة الميثان CH_4 (بالجرامات) التي يجب حرقها لإنتاج **3564 kJ** من الحرارة؟

(الكتلة المولية لـ CH_4 =16 g/mol)

Substance المادة	Chemical formula الصيغة الكيميائية	$\Delta H^{\circ}_{comb}(kJ/mol)$
Methane الميثان	$CH_4(g)$	−891

A. 4 g

B. 64 g

C. 0.25 g

D. 16 g

A fuel releases 1684.8 kJ of heat when 0.600 mol of it is burned. Which of the substances listed in the following table represents this fuel?

يُطلق وقود 1684.8 kJ من الحرارة عند احتراق 0.600 mol منه. أي أنواع المواد المُدرجة في الجدول التالي تُمثل هذا الوقود؟

Substance المادة	chemical formula الصيغة الكيميائية	$\Delta H^{\circ}_{comb}(kJ/mol)$
Sucrose السكروز	$C_{12}H_{22}O_{11}(s)$	−5644
Octane الأوكتان	$C_8H_{18}(l)$	−5471
Glucose الجلوكوز	$C_6H_{12}O_6(s)$	−2808
Propane البروبان	$C_3H_8(g)$	−2219

A. Sucrose

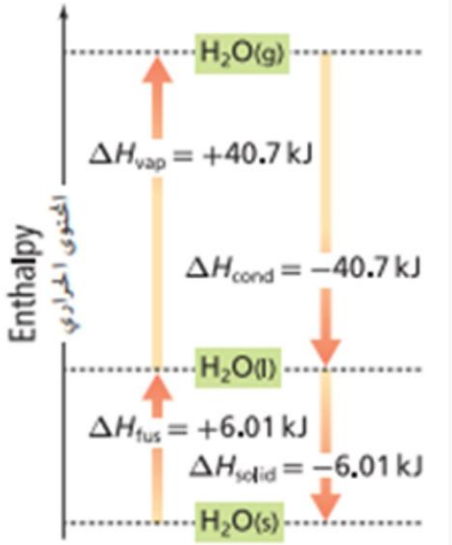
B. Octane

C. Glucose

D. Propane

As you step out of a hot shower, you shiver as heat is taken from your skin. Using the figure below, what is the symbol represents this heat?

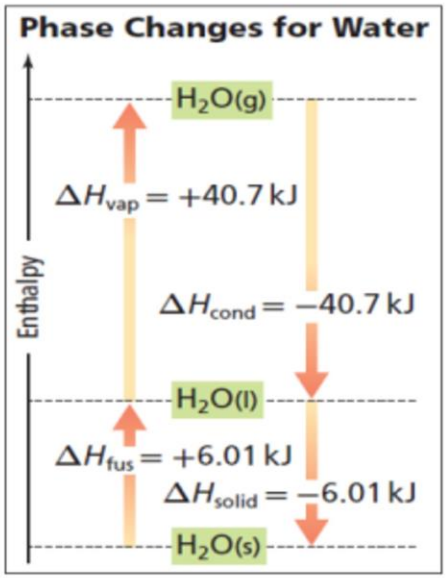
عند خروجك من حمام ساخن، سوف ترتعش عندما تخرج الحرارة من جلدك . موظفا الشكل أدناه ، ما الرمز الذي يمثل هذه الحرارة ؟



A. ΔH_{vap}
B. ΔH_{fus}
C. ΔH_{solid}
D. ΔH_{cond}

Which of the following statement is **incorrect** according to the diagram shown?

أي من العبارات **غير صحيحة** اعتمادا على الشكل



A. The system energy increase as ice melts and then evaporates

تزداد طاقة النظام عند انصهار الثلج ثم تبخره

B. The system energy increase as water vapor condenses and then solidifies

تزيد طاقة النظام عند تكثيف بخار الماء ثم تجمده

C.The vaporization of water and melting of ice are considered as endothermic process

تعتبر عمليات تبخر الماء وصهر الثلج عمليات ماصة للحرارة.

D. The condensation and solidification of water are considered as exothermic process

تعتبر عمليات تكثف الماء وتجمده طارده للحرارة

Which of the following changes is an endothermic? أي التغيرات التالية ماصة للحرارة؟

I	Water vaporization
II	Melting of ice
III	Water freezing
IV	Condensing of water vapor

I	تبخير الماء
II	انصهار الثلج
III	تجمد الماء
IV	تكثف بخار الماء

A. I and II / ا و II

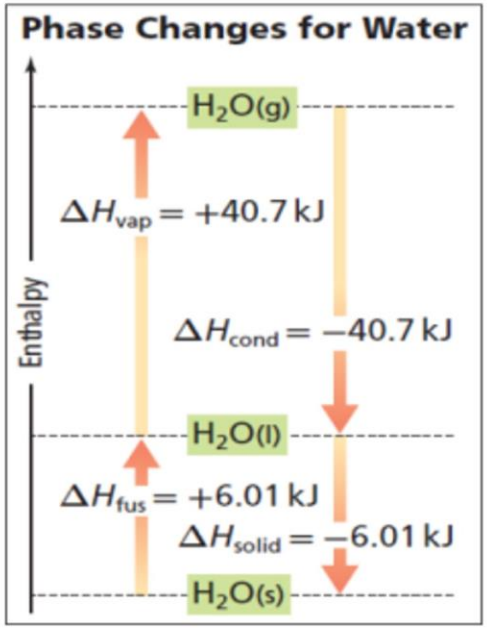
B. I and III / ا و III

C. II and IV / II و IV

D. III and IV / III و IV

calculate how much heat is required to vaporize 5.33 mol of water at 100°C.

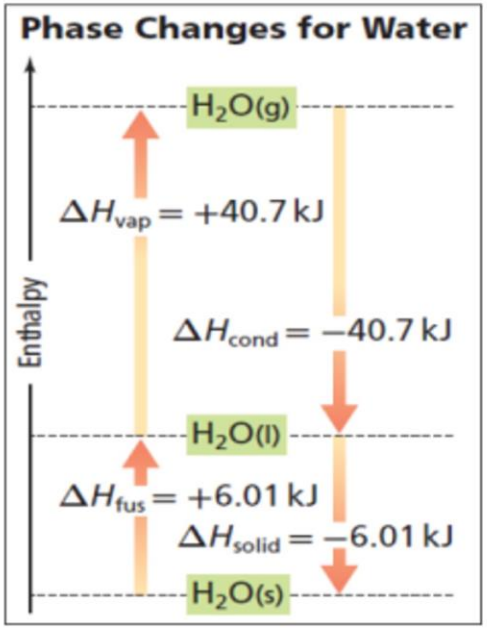
احسبي كم كمية الحرارة اللزمة لتبخير 5.33 mol من الماء عند 100°C. ؟



A. 9.4 kJ
B. 26.02 kJ
C. 217 kJ
D. 176 kJ

What is the heat released in freezing 4.5g water? (molar mass of water = 18.02 g/mol)

ما مقدار الحرارة الناتجة عند تجمد 4.5g من الماء ؟



A. 1.5 kJ
B. 27.04 kJ
C. 10.16 kJ
D. 183.15 kJ

CHM.5.5.02.002.01 Calculate, using Hess's law, the ΔH of a reaction

Using Hess's law what is ΔH°_{rxn} for the reaction?

مستخدمًا قانون هس، ما مقدار ΔH°_{rxn} للتفاعل التالي؟

$$Pb_{(s)} + PbO_{2(s)} + 2SO_{3(g)} \rightarrow 2PbSO_{4(s)} \qquad \Delta H^{\circ}_{rxn} = ?$$

given the following reactions:

إذا علمت التفاعلات التالية:

$$SO_{3\,(g)} + H_2O_{(l)} \rightarrow H_2SO_{4(l)} \qquad \Delta H° = -133\ kJ$$

$$Pb_{(s)} + PbO_{2(s)} + 2H_2SO_{4(l)} \rightarrow 2PbSO_{4(s)} + 2H_2O_{(l)} \qquad \Delta H° = -509\ kJ$$

A. +642 kJ

B. -775 kJ

C. -376 kJ

D. +376 kJ

Using Hess's law and the change in enthalpy for the following two reactions :

مستخدما قانون هس والتغيرات في المحتوى الحراري للتفاعلين التاليين :

$2A + \frac{3}{2}C_2 \rightarrow A_2C_3 \qquad \Delta H = -1874\ kJ$	1
$2B + \frac{3}{2}C_2 \rightarrow B_2C_3 \qquad \Delta H = -285\ kJ$	2

What is ΔH for the following reaction?

ما قيمة ΔH *للتفاعل التالي*

$$2A + B_2C_3 \rightarrow 2B + A_2C_3 \qquad \Delta H = ?$$

A. -1589 kJ

B. −1222 kJ

C. -2159 kJ

D. +2159 kJ

CHM.5.5.02.002.01 Calculate, using Hess's law, the ΔH of a reaction

Use Hess's Law to calculate the energy change for the reaction that produces SO_3.	مستخدمًا قانون هس ، ما التغير في الطاقة للتفاعل الذي يُنتج SO_3؟
$2S(s) + 2O_2(g) \rightarrow 2SO_2(g)$ $\Delta H = -594kJ$ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$ $\Delta H = -198kJ$	
A. ΔH = +406KJ	
B. ΔH = -406KJ	
C. ΔH = +792kJ	
D. ΔH = -792kJ	

Use Hess's law to determine ΔH for the reaction $\mathbf{NO(g) + O(g) \rightarrow NO_2(g)}$ $\Delta H = ?$ given the following reactions.

استخدمي قانون هس لحساب ΔH للتفاعل $\mathbf{NO(g) + O(g) \rightarrow NO_2(g)}$ $\Delta H = ?$ من التفاعلات التالية

a. $O_2(g) \rightarrow 2O(g)$ $\Delta H = +495$ kJ

b. $2O_3(g) \rightarrow 3O_2(g)$ $\Delta H = -427$ kJ

c. $NO(g) + O_3(g) \rightarrow NO_2(g) + O_2(g)$ $\Delta H = -199$ kJ

A. 233 kJ
B. -131 kJ
C. -233 kJ
D. 131 kJ

CHM.5.5.02.002.01 Calculate, using Hess's law, the ΔH of a reaction

Using Hess's law and the changes in enthalpy for the following reactions:

مُستخدمًا قانون هس والتغيرات في المحتوى الحراري للتفاعلات التالية :

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)} \qquad \Delta H = -393.5\ kJ$$

$$H_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow H_2O_{(l)} \qquad \Delta H = -285.8\ kJ$$

$$CH_{4(g)} + 2O_{2(g)} \rightarrow CO_{2(g)} + 2H_2O_{(l)} \qquad \Delta H = -890.8\ kJ$$

What is ΔH_f° for the following reaction?

ما قيمة ΔH_f° للتفاعل التالي؟

$$C_{(s)} + 2H_{2(g)} \rightarrow CH_{4(g)} \qquad \Delta H_f^\circ = ??$$

A. -74.3 KJ/mol

B. -212.3 KJ/mol

C. -181.2 KJ/mol

D. -233.4 KJ/mol

Challenge Two enthalpy of formation equations, **a** and **b**, combine to form the equation for the reaction of nitrogen oxide and oxygen. The product of the reaction is nitrogen dioxide: $NO_{(g)} + \frac{1}{2}O_{2\,(g)} \rightarrow NO_{2\,(g)}$ $\Delta H = -58.1$ kJ

a. $\frac{1}{2}N_{2(g)} + \frac{1}{2}O_{2(g)} \rightarrow NO_{(g)}$ $\Delta H°_f = -91.3$ kJ

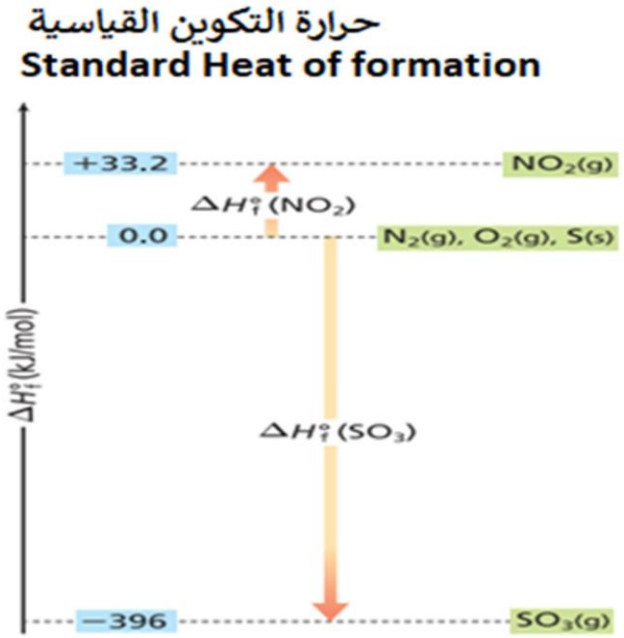
b. $\frac{1}{2}N_{2(g)} + O_{2(g)} \rightarrow NO_{2(g)}$ $\Delta H°_f = ?$

What is $\Delta H°_f$ for equation **b**?

A. +33.2 kJ
B. −149.4 kJ
C. -33.2 kJ
D. 149.4 kJ

Using information from the figure below, which of the following statements is correct ?

مستخدما بيانات الشكل أدناه، أي العبارات التالية صحيحة ؟



A. Both the reaction of formation of $NO_{2(g)}$ and formation of $SO_{3(g)}$

كلا من تفاعلات تكوين $NO_{2(g)}$ وتكوين $SO_{3(g)}$ تعد ماصة للحرارة

B. 33.2 kJ is released to form 1 mole of $NO_{2(g)}$

يتحرر 33.2 kJ عند تكون مول واحد من $NO_{2(g)}$

C. 33.2 kJ is absorbed to form 1 mole of $SO_{3(g)}$

يتم امتصاص 33.2 kJ عند تكون مول واحد من $SO_{3(g)}$

D. the formation of $NO_{2(g)}$ is an endothermic, while the formation of $SO_{3(g)}$ is an exothermic .

يعد تفاعل تكوين $NO_{2(g)}$ ماصا للحرارة بينما تفاعل تكوين $SO_{3(g)}$ طاردا للحرارة

Which of the following equations represented a thermochemical equation for formation of one mol of **NO_2** from its elements in their standard states ?

أي المعادلات التالية تمثل المعادلة الكيمائية الحرارية لتكوين واحد مول من **NO_2** من عناصره في حالتها القياسية ؟

A. $NO_{(g)} + \frac{1}{2} O_{2\,(g)} \rightarrow NO_{2\,(g)}$ ΔH = -58.1 kJ

B. $\frac{1}{2} N_{2(g)} + O_{2(g)} \rightarrow NO_{2(g)}$ ΔH = +33.2 kJ

C. $\frac{1}{2} N_{2(g)} + \frac{1}{2} O_{2(g)} \rightarrow NO_{(g)}$ ΔH = +91.3 kJ

D. $\frac{3}{2} N_{2(g)} + 3O_{2(g)} \rightarrow 3NO_{2(g)}$ ΔH = +99.6 kJ

Which Reaction represent the standard enthalpies of formation for compound ?

أي يتفاعل يمثل حرارة تكوين قياسية للمركب الناتج من التفاعلات التالية ؟

A. $NO_{(g)} + \frac{1}{2} O_{2\,(g)} \rightarrow NO_{2\,(g)}$ ΔH = -58.1 kJ

B. $S_{(s)} + \frac{3}{2} O_{2\,(g)} \rightarrow SO_{3\,(g)}$ ΔH = -396 kJ

C. $SO_{2(g)} + 2H_2O_{(l)} \rightarrow H_2SO_{4(l)}$ ΔH = -814 kJ

D. $H_{2(g)} + F_{2(g)} \rightarrow 2HF_{(g)}$ ΔH = -546 kJ

Using standard enthalpies of formation what is the ΔH°_{rxn} for the combustion of methane?	مستخدمًا حرارة التكوين القياسية ما ΔH°_{rxn} للتفاعل احتراق الميثان؟
$CH4_{(g)} + 2O2_{(g)} \rightarrow CO2_{(g)} + 2H2O_{(l)}$ ΔH° f (CO2) = -394 kJ ΔH° f (H2O) = -286 kJ ΔH° f (CH4) = -75 kJ	
A. -605 kJ	
B. -640 kJ	
C. -891 kJ	
D. -1041 kJ	

Apply Phosphorous trichloride is a starting material for the preparation of organic phosphorous compounds. Demonstrate how thermochemical equations **a** and **b** may be used to determine the enthalpy change for the reaction described by the equation $PCl_3(l) + Cl_2(g) \rightarrow PCl_5(s)$.

a. $P_4(s) + 6Cl_2(g) \rightarrow 4PCl_3(l)$ ΔH = -1280 kJ

b. $P_4(s) + 10Cl_2(g) \rightarrow 4PCl_5(s)$ ΔH = -1774 kJ

A. +124 kJ
B. −149.4 kJ
C. -124 kJkJ
D. 149.4 kJ

Using standard enthalpies of formation table below

What is the ΔH°_{rxn} value for the following reaction?

مُستخدمًا جدول قيم حرارة التكوين القياسية أدناه

ما قيمة ΔH°_{rxn} للتفاعل التالي؟

$$4NH_{3(g)} + 7O_{2(g)} \rightarrow 4NO_{2(g)} + 6H_2O_{(l)}$$

المادةSubstance	ΔH°_{f} (kJ/mol)
$NH_{3(g)}$	-46.19
$H_2O_{(l)}$	-285.8
$NO_{2(g)}$	+33.2

A. -1397 kJ

B. + 1797 kJ

C. -1797 kJkJ

D. +299 kJ

What is the enthalpy change ΔH *for the reaction below*

ما قيمة التغير في المحتوى الحراري للتفاعل ΔH أدناه ؟

$$2H_2S_{(g)} + 3O_{2(g)} \longrightarrow 2H_2O_{(l)} + 2SO_{2(g)} \quad , \Delta H = ??$$

Use the following equations

مستخدما المعادلات التالية :

$$S_{(s)} + O_{2(g)} \longrightarrow SO_{2(g)} \quad , \Delta H^0_f = -296.8 \text{ kJ/mol}$$

$$H_{2(g)} + \frac{1}{2} O_{2(g)} \longrightarrow H_2O_{(l)} \quad , \Delta H^0_f = -285.8 \text{ kJ/mol}$$

$$H_{2(g)} + S_{(s)} \longrightarrow H_2S_{(g)} \quad , \Delta H^0_f = -20.6 \text{ kJ/mol}$$

A. -603.2 kJ
B. −562.0 kJ
C. -1206.4 kJ
D. -1124 kJ

Using standard enthalpies of formation below, what is the ΔH°_{rxn} value for the following reaction ?

مُستخدمًا قيم حرارة التكوين القياسية أدناه ، ما قيمة ΔH°_{rxn} للتفاعل التالي؟

$$2ZnS_{(s)} + 3O_{2(g)} \rightarrow 2ZnO_{(s)} + 2SO_{2(g)}$$

$SO_{2(g)}$	$ZnO_{(s)}$	$ZnS_{(s)}$	substance المادة
-296.8	-348.3	-206.0	(kJ/mol) ΔH°_{f}

A. -901.3 KJ

B. -270.6 KJ

C. -593.7 KJ

D. -878.2 KJ

مؤسسة الإمارات للتعليم المدرسي
EMIRATES SCHOOLS ESTABLISHMENT

Unit:

Reactions Rates



11A

Chemistry teacher : Sameera Albreiki

In the reaction : $2HI_{(g)} \rightarrow I_{2(g)} + H_{2(g)}$ the concentration of HI is 0.045 M at the beginning of the reaction . At 3.00 s the concertation of HI is 0.03 M. What is the average reaction rate over the given time period ?
A. + 0.005 mol/L.S
B. - 0.005 mol/L.S
C. + 0.015 mol/L.S
D. - 0.015 mol/L.S

In the gas –phase reaction : $I_{2(g)} + Cl_{2(g)} \rightarrow 2ICl$ [I2] changes from 0.4 M at 0.0 s to 0.3M at 4.00s. What is the average reaction rate over the given time period expressed in mol/L.s ?
A. 0.025
B. 0.035
C. 0.870
D. 0.690

Using the experimental data in the following table, if the average reaction rate for the reaction ,expressed in moles of NO formed, is 0.05 mol/L.s . What concentration of NO would be present after 0.1 s?

$$N_2(g) + O_2(g) \rightarrow 2NO(g)$$

$[NO]$ (M)	$[O_2]$ (M)	$[N_2]$ (M)	Time (s) الزمن
0.000	0.300	0.500	0.00
	0.250	0.450	0.100

A. 0.02

B. 0.5

C. 0.2

D. 0.005

Using the experimental data in the following table, if the average reaction rate for the reaction ,expressed in moles of NO formed, is 0.05 mol/L.s . What concentration of NO would be present after 0.3s?

$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$			
time	$[N_2](M)$	$[O_2](M)$	$[NO](M)$
0.00	0.50	0.30	0.00
0.30	0.35	0.15	

A. 0.02

B. 0.5

C. 0.015

D. 0.005

Using the experimental data in the following table, if the average reaction rate for the reaction ,expressed in moles of NO formed, is 0.05 mol/L.s . What concentration of NO would be present after 0.5s?

$N_{2(g)} + O_{2(g)} \rightarrow 2NO_{(g)}$

time	$[N_2](M)$	$[O_2](M)$	$[NO](M)$
0.00	0.50	0.30	0.00
0.50	0.25	0.05	

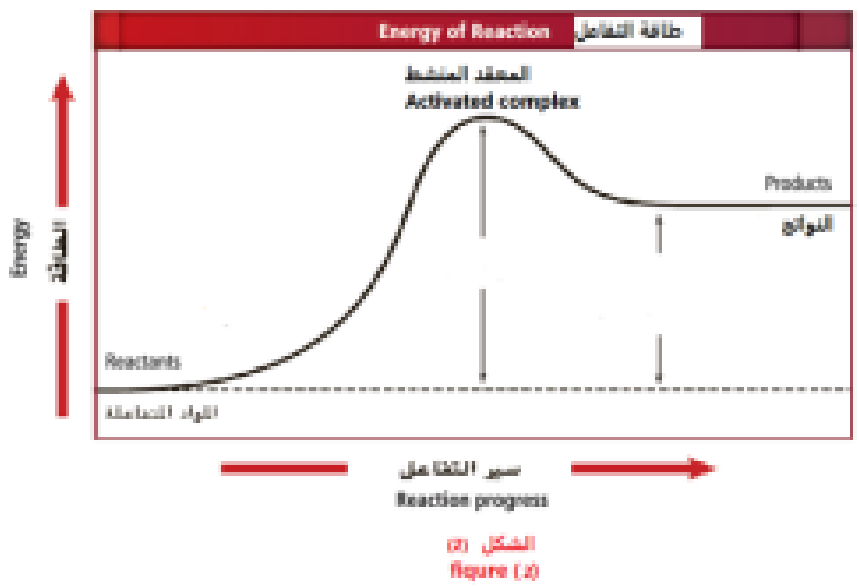
A. 0.02

B. 0.025

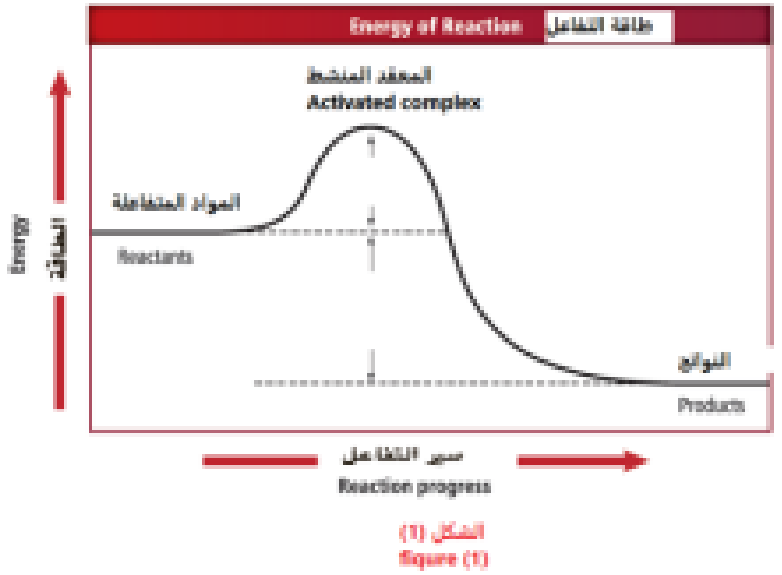
C. 0.015

D. 0.005

Using the two figures below , which of the following is correct?



الشكل (2)
figure (2)



الشكل (1)
figure (1)

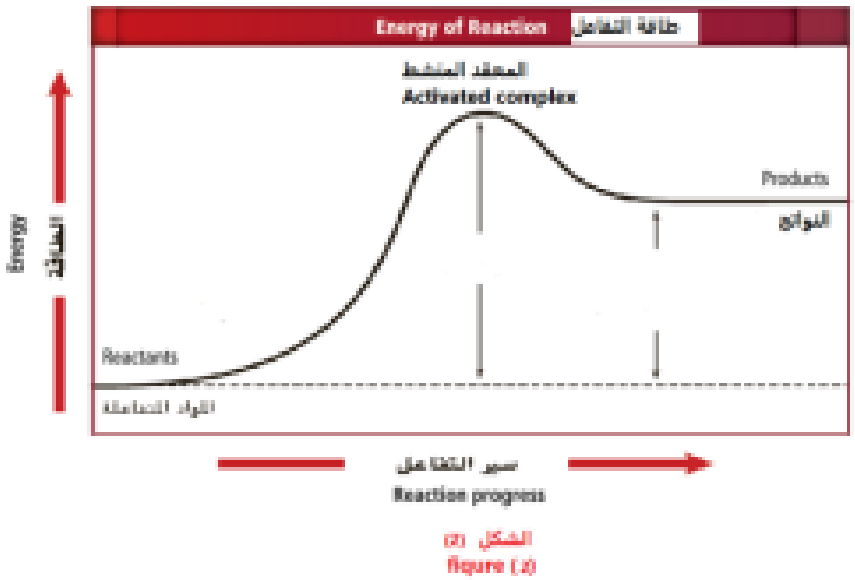
A. Energy is released in figure (1) while energy is absorbed in figure (2)

B. Energy is released in figure (2) while energy is absorbed in figure (1)

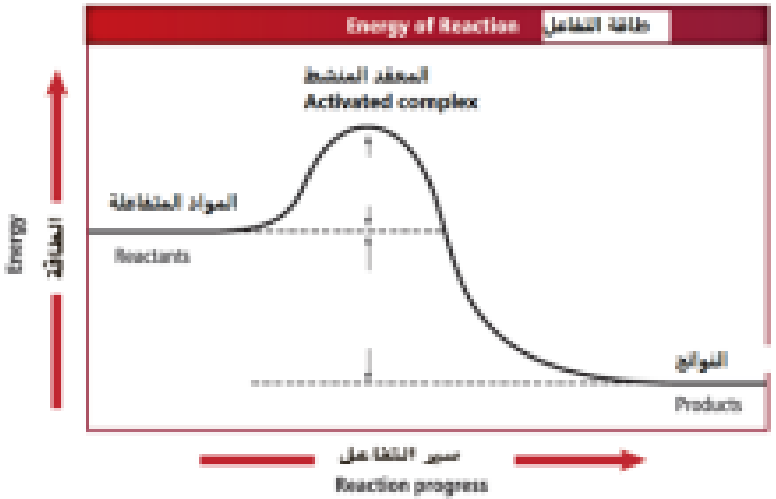
C. The activation energy in figure (1) is greater than the activation energy in figure (2)

D. The activation energy in figure (1) equals the activation energy in figure (1)

Using the two figures below , which of the following is correct?



الشكل (2)
figure (2)



الشكل (1)
figure (1)

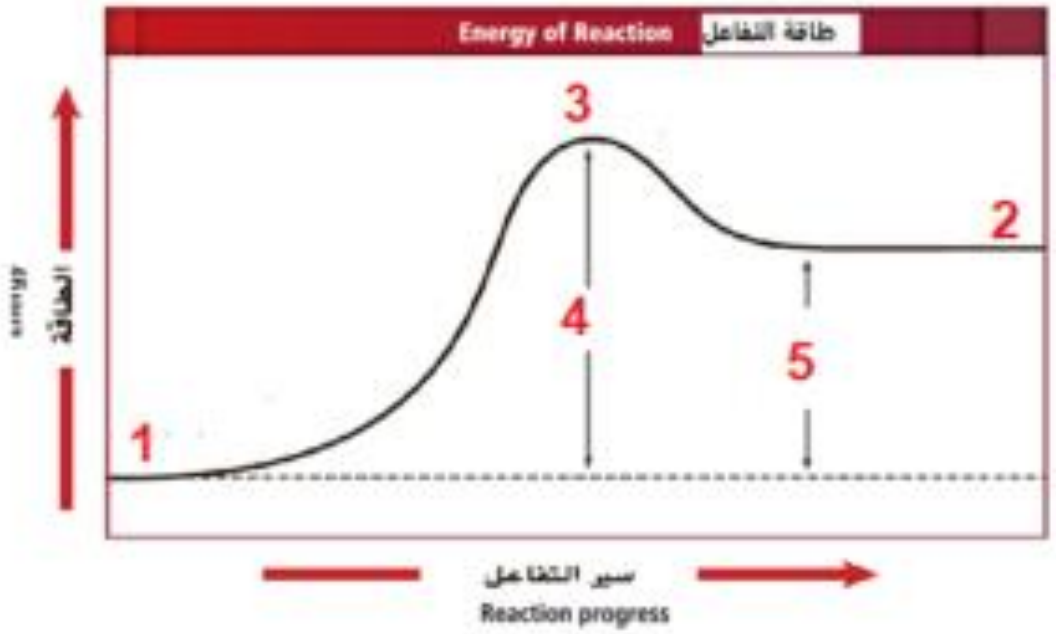
A. Energy is released in figure (2) while energy is absorbed in figure (1)

B. The activation energy in figure (2) is greater than the activation energy in figure (1)

C. The activation energy in figure (1) is greater than the activation energy in figure (2)

D. The activation energy in figure (1) equals the activation energy in figure (1)

Regarding the figures below , which of the following is correct?



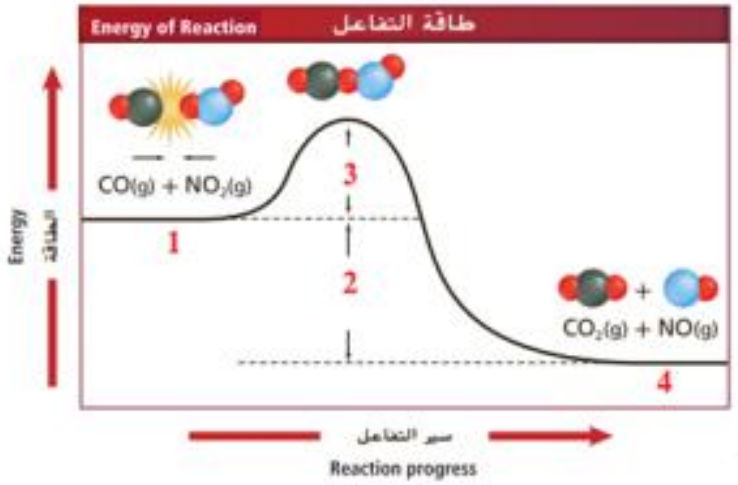
A. **1** indicates products while **2** indicates reactants.

B. **4** indicates activation energy while **5** indicates energy absorbed by reaction.

C. **4** indicates activation energy while **5** indicates energy released by reaction.

D. **1** indicates activated complex while **2** indicates reactants.

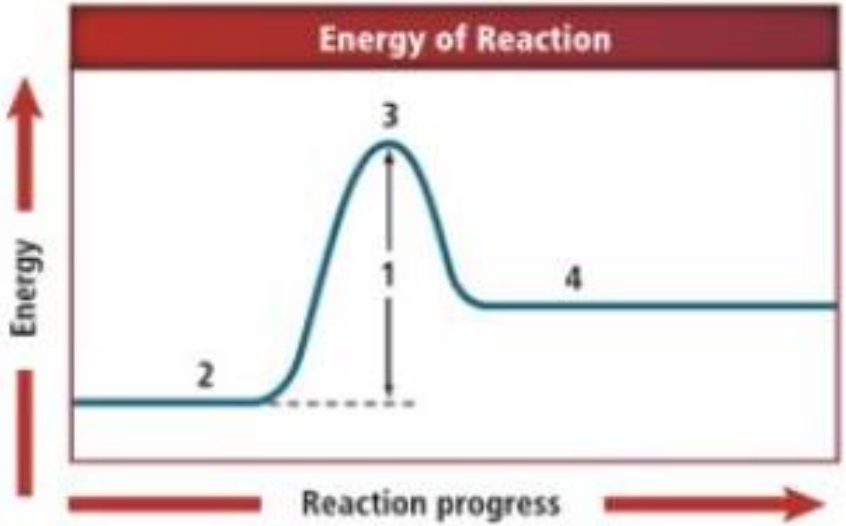
Represented by the graph below?



	1	2	3	4
A	**Reactants**	**Energy released by reaction**	**Activation Energy**	**Products**
B	**Reactants**	**Activation Energy**	**Energy released by reaction**	**Products**
C	**Reactants**	**Products**	**Activation Energy**	**Energy released by reaction**
D	**Activation Energy**	**Reactants**	**Products**	**Energy released by reaction**

A
B
C
D

Which of the following is represented by the number 3 in the figure below ?



A. Activation energy

B. Activation complex

C. Reactants

D. Products

Calculate the average rate of a reaction between hypothetical molecules A and B if the concentration of A changes from 1.00M to 0.50M in 2.00 s.

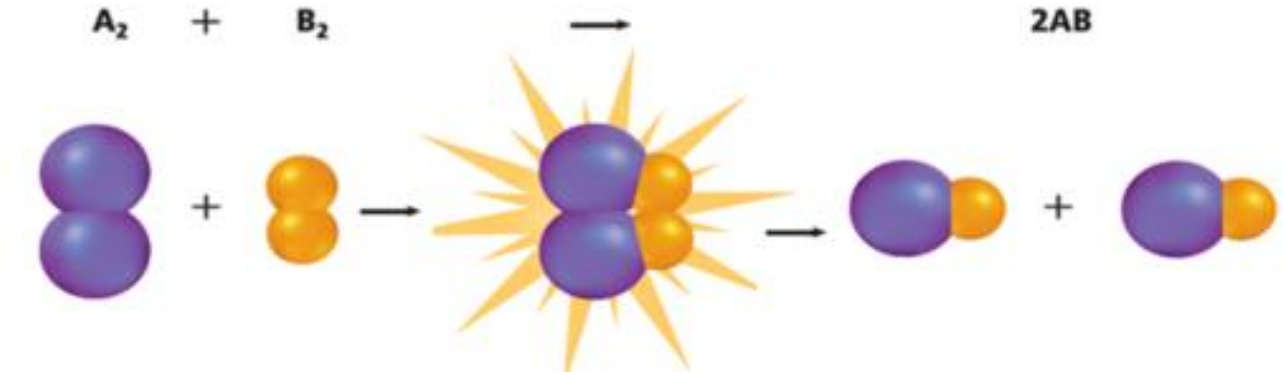
A. 0.25 mol/(L.s)

B. 0.75 mol/(L.s)

C. 2.00 mol/(L.s)

D. 4 mol/(L.s)

Why the collisions between the reactant molecules success to produce products in the following reaction ?



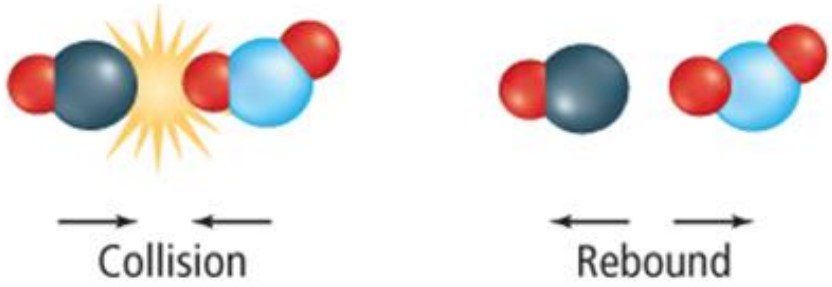
A. Because the molecules collide in the correct orientation, and they have sufficient energy to react.

B. Because the molecules collide in the incorrect orientation, and they have insufficient energy to react.

C. Because the molecules collide in the correct orientation, and they have insufficient energy to react.

D. Because the molecules collide in the incorrect orientation, and they have sufficient energy to react.

Why the collisions between the reactant molecules not success to produce products in the following reaction ?



A. Because the molecules collide in the correct orientation, and they have sufficient energy to react.

B. Because the molecules collide in the incorrect orientation, and they have insufficient energy to react.

C. Because the molecules collide in the correct orientation, and they have insufficient energy to react.

D. Because the molecules collide in the incorrect orientation, and they have sufficient energy to react.

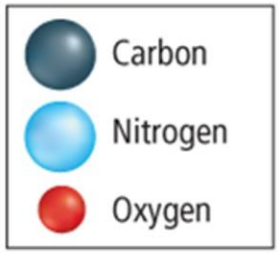
CHM.5.4.01.002.03 List the two conditions, according to the collision theory, that must be met for a collision between reactant molecules to be effective in producing new chemical species

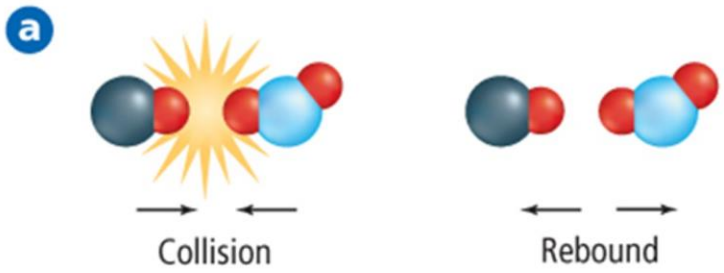
Textbook + Figures 3, 4

50, 51

Why the collisions between the reactant molecules not success to produce products in the following reaction ?

$CO\ (g) + NO_2\ (g) \rightarrow CO_2\ (g) + NO\ (g)$





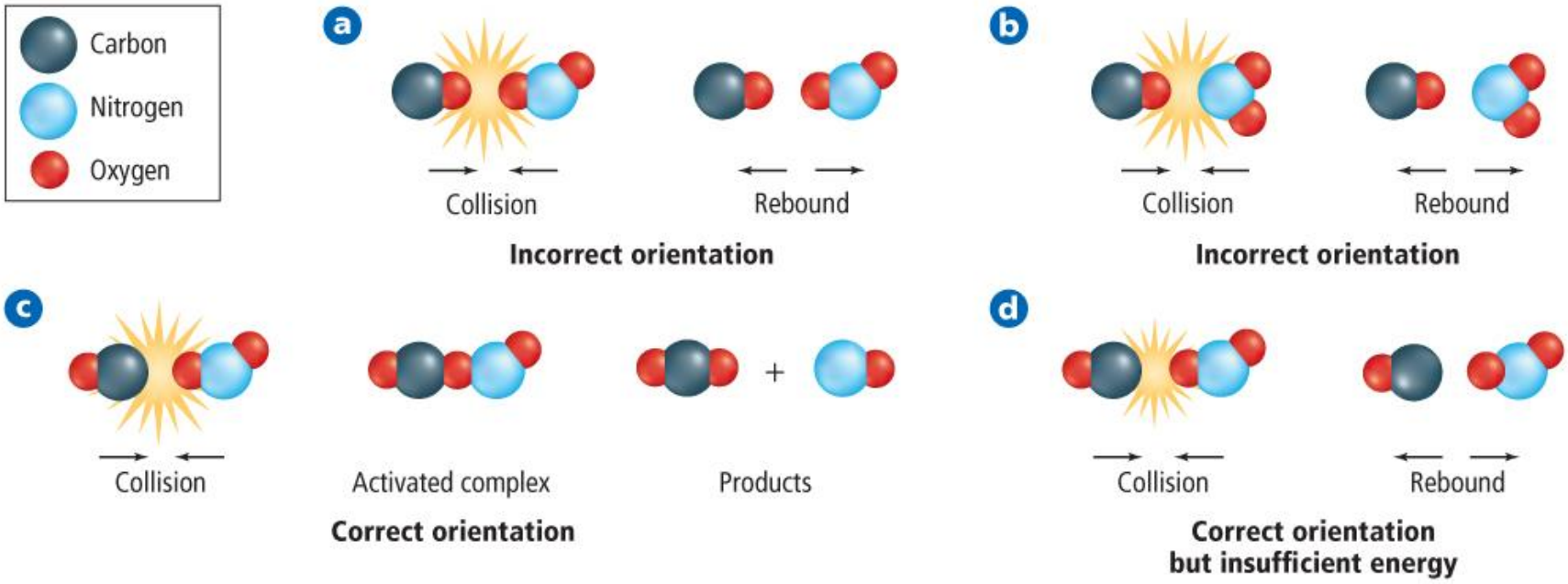
A. Because the molecules collide in the correct orientation, and they have sufficient energy to react.

B. Because the molecules collide in the incorrect orientation, and they have insufficient energy to react.

C. Because the molecules collide in the correct orientation, and they have insufficient energy to react.

D. Because the molecules collide in the incorrect orientation, and they have sufficient energy to react.

Why the collision produce a chemical reaction in figure (c) below?



A. sufficient energy

B. insufficient energy

C. Correct orientation

D. A and C

The rate law for the reaction between nitrogen monoxide and hydrogen is **Rate = $k[NO_2]^2[H_2]$**. What happens to the rate if the concentration of hydrogen (H_2) doubles?
A. The rate doubles
B. The rate triples.
C. The rate quadruples.
D. The rate is reduced by half.

The rate law for the following reaction: $aA + bB \rightarrow cC + dD$ is $Rate = k[A]^2[B]^1$ What happens to the rate if the concentration of B doubles ?
A. The rate reduces by half.
B. The rate increases four times.
C. The rate increases three times.
D. The rate doubles.

The rate law for the following reaction: $aA + bB \rightarrow cC + dD$ is $Rate = k[A]^2[B]^1$ What happens to the rate if the concentration of A doubles ?
A. The rate reduces by half.
B. The rate increases four times.
C. The rate increases three times.
D. The rate doubles.

The chemical reaction: $2NO_{(g)} + O_2 \rightarrow 2NO_{2(g)}$ Has the following rate law $Rate = k[NO]^2[O_2]$ If the concentration of NO decreases by half, what happens to the reaction rate?
A. Reduces to one-half.
B. Becomes double
C. Reduces to one-quarter.
D. Remains the same .

The reaction below is second order in O_2 and fourth order overall , what is the rate law for the reaction ? $2SO_{2(g)} + O_2 \rightarrow 2SO_{3(g)}$
A. Rate = $k [SO_2]^3[O_2]$
B. Rate = $k [O_2]^4[SO_2]$
C. Rate = $k [O_2]^2[SO_2]^2$
D. Rate = $k [SO_2]^4[O_2]^2$

The rate law for the following reaction: $aA + bB \rightarrow cC + dD$ is Rate = $k [A] [B]^2$ What happens to the rate if the concentration of B doubles ?
A. The rate reduces by half.
B. The rate increases four times.
C. The rate increases three times.
D. The rate doubles.

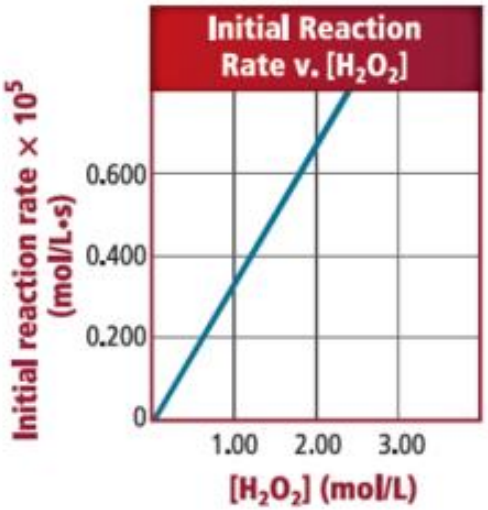
What is the unit of the specific rate constant k for a reaction that has the rate law: $Rate = k[A]^2$
A. $L^2/(mol^2.s)$
B. L / (mol.s)
C. s^{-1}
D. s

What is the unit of the specific rate constant k for a reaction that has the rate law: $Rate = k[A]^3$
A. $L^2/(mol^2.s)$
B. L / (mol.s)
C. s^{-1}
D. s

Decomposition reaction of H_2O_2 :

$$2H_2O_{2(aq)} \rightarrow 2H_2O_{(aq)} + O_{2(g)}$$

Is first order and the graph below shows relationship between the concentration and its rate decomposition, when $[H_2O_2]$ equals 1.5 mol/L . Which of the following is the correct unit of specific rate constant **k** ?



A. L / (mol.s)
B. L^2/ (mol^2.s)
C. s^{-1}
D. s

Given the following experimental data, use the method of initial rates to determine the rate law for the reaction : aA + bB → products.

Experimental Initial Rates for *a*A + *b*B → products			
Trial	**Initial [A] (*M*)**	**Initial [B] (*M*)**	**Initial Rate (mol/L·s)**
1	0.30	0.10	1.20×10^{-2}
2	0.60	0.10	4.80×10^{-2}
3	0.60	0.20	9.60×10^{-2}

A. Rate = k [A] $[B]^2$

B. Rate = k $[A]^2$ $[B]^2$

C. Rate = k $[A]^2$ [B]

D. Rate = k [A] [B]

Using the experimental data in the following table.
What is the rate law for the reaction : aA + bB → products.

aA + bB → products			
التجربة Trial	التركيز الابتدائي Initial concentration [A] (M)	التركيز الابتدائي Initial concentration [B] (M)	السرعة الابتدائية Initial Rate (mol/(L.s))
1	0.273	0.273	3.00×10^{-3}
2	0.546	0.273	3.00×10^{-3}
3	0.546	0.546	6.00×10^{-3}

A. Rate = k [A]

B. Rate = k [A] [B]

C. Rate = k $[A]^2$ [B]

D. Rate = k [B]

Using the experimental data in the following table.
What is the rate law for the reaction : $aA + bB \rightarrow$ products.

Trail	Initial concentration [A] (M)	Initial concentration [B] (M)	Initial Rate (mol/L.s)
1	0.273	0.273	3.00×10^{-3}
2	0.546	0.273	1.20×10^{-2}
3	0.546	0.546	9.6×10^{-2}

A. Rate = k [A]

B. Rate = k [A] [B]

C. Rate = k $[A]^2$ $[B]^3$

D. Rate = k[A] $[B]^2$

Using the experimental data in the following table.
What is the rate law for the reaction : $aA + bB \rightarrow$ products.

Trail	Initial concentration [A] (M)	Initial concentration [B] (M)	Initial Rate (mol/L.s)
1	0.273	0.273	3.00×10^{-3}
2	0.273	0.546	1.20×10^{-2}
3	0.546	0.546	9.6×10^{-2}

A. Rate = k [A]

B. Rate = k [A] [B]

C. Rate = k $[A]^2$ $[B]^3$

D. Rate = k$[A]^3$ $[B]^2$

Using the experimental data in the following table.
What is the rate law for the reaction.

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

التجربة Trial	التركيز الابتدائي Initial concentration [NO] (M)	التركيز الابتدائي Initial concentration $[O_2]$ (M)	السرعة الابتدائية Initial Rate (mol/(L.s))
1	0.030	0.020	0.0041
2	0.060	0.020	0.0164
3	0.060	0.040	0.0328

A. Rate = k $[NO]^2[O_2]$

B. Rate = k $[NO]^2[O_2]^2$

C. Rate = k $[NO][O_2]^2$

D. Rate = k $[NO][O_2]$

Using the experimental data in the following table to calculate the value of the specific rate constant, k. The rate law for this reaction is Rate = k $[NO]^2[O_2]$

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

التجربة Trial	التركيز الابتدائي Initial concentration [NO] (M)	التركيز الابتدائي Initial concentration $[O_2]$ (M)	السرعة الابتدائية Initial Rate (mol/(L.s))
1	0.030	0.020	0.0041
2	0.060	0.020	0.0164
3	0.060	0.040	0.0328

A. 2.3×10^2 $L^3/mol^2.s$

B. 6.83 $L^3/mol^2.s$

C. 3.4×10^2 $L^3/mol^2.s$

D. 4.4×10^{-5} $L^3/mol^2.s$

Using the experimental data in the following table to calculate the value of the specific rate constant, k. The rate law for this reaction is Rate = k [A]^2[B]

Experimental Initial Rates for *a*A + *b*B → products

Trial	Initial [A] (*M*)	Initial [B] (*M*)	Initial Rate (mol/L·s)
1	0.30	0.10	1.20×10^{-2}
2	0.60	0.10	4.80×10^{-2}
3	0.60	0.20	9.60×10^{-2}

A. 2.3×10^2 $L^3/mol^2.s$

B. 0.4 $L^3/mol^2.s$

C. 1.33 $L^3/mol^2.s$

D. 3.6×10^{-5} $L^3/mol^2.s$

The reaction between A and B is first order in A and B and the specific rate constant (k) =0.500 L/(mol.s). calculate the instantaneous rate when the reactants concentration are [A]= 2×10^{-2} M and [B] = 1.5×10^{-2} M
A. 1.5×10^{-4} mol/(L.s)
B. 4.8×10^{-4} mol/(L.s)
C. 6×10^{-4}mol/(L.s)
D. 3×10^{-2} mol/(L.s)

The following reaction is first order in H_2 and second order in NO with rate constant of 2.90×10^{2} $L^2/(mol^2.s)$ $2NO_{(g)} + H_{2(g)} \rightarrow N_2O_{(g)} + H_2O$ calculate the instantaneous rate when the reactants concentration are [NO]= 0.005 M and [H_2] = 0.002 M
A. 1×10^{-5} mol/(L.s)
B. 4.8×10^{-4} mol/(L.s)
C. 1.45×10^{-5}mol/(L.s)
D. 1.3×10^{-2} mol/(L.s)

The following reaction is first order in H_2 and second order in NO with rate constant of $2.90 \times 10^2\ L^2/(mol^2.s)$ $2NO_{(g)} + H_{2(g)} \rightarrow N_2O_{(g)} + H_2O$ Calculate [NO] concentration for reaction if the rate is are 9.00×10^{-5} mol/(L.s) and $[H_2] = 0.003$ M
A. 1.03×10^{-4} mol/(L.s)
B. 4.8×10^{-4} mol/(L.s)
C. 1.02×10^{-2} mol/(L.s)
D. 1.3×10^{-2} mol/(L.s)

إن الله دائمًا يُحقق
المستحيلات
بالطريقة الأكثر إستحالة
فاطمئن

